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CSEN122 Computer Architecture Project

Abstract

My CPU implements 11-instructions based on a derivation of the RISC-V (32-bit) CPU architecture called SCU ISA. The SCU ISA architecture will use word addressing.
For my architecture, I decided not to implement the Forwarding and Hazard Detection Unit, and instead opted to hardcode NOP instruction into my assembly code.

The CPU is using the 5-stage pipelined architecture:

1. IF: Instruction Fetch
2. ID: Instruction Decode / Register Read
3. EX: Execute
4. MEM: Data Memory
5. WB: Write-Back

And I are using 4 pipeline buffers:

1. IF/ID: 64 bits
2. ID/EX: 145 bits
3. EX/MEM: 144 bits
4. MEM/WB: 105 bits

The buffer bit-widths are calculated in Table 5 of my Appendix section.

The datapath diagram I designed shows the full flow of these signals betIen these 5 components. Because my design does not implement forwarding or hazard detection, I inserted NOPs in the assembly program to avoid hazards.

I achieved adding these NOP instructions by assessing the code for read-after-write (RAR) dependencies.
For my simulation I chose to test the professor provided stencil code **`CSEN122-F25-SCU_ISA_Assembly_1d_Stencil.txt`**.

I decided to write a JavaScript Program (Code 2) that converts the asm code into binary so I can easily paste it into **`inst_mem.v`**.

CPU Design Description:

Instruction Fetch (IF) Stage 1

The PC uses a 32-bit register and is updated every cycle. A branch or jump address could also influence the changes of the PC address.

This stage has the following components:

- Instruction Memory
- PC
- adder: computing PC + 1 (word-addressing)
- mux: (logically implemented using conditionals) two connected together.
  Each mux choosing betIen:
  1. Signal: PCSrc; 0 - rs_data_EX; 1 - rs_data_MEM;
  2. Signal: J ; 0 – (PC + 1) ; 1 – 1st mux

PCSrc is comprised of go_branch and Jump (J) signal.
Refer to Data Memory (MEM) Stage 4 for a description of go_branch.

Instruction Decode (ID) Stage 2

This stage contains the following components:

- Register File
- Immediate Generator
- Control Unit

Register File

The register file contains 64 register by 32 bits. The design uses synchronous write and read on the positive edge.

Immediate Generator

The immediate field (10-bit signed) is extracted from the instruction [31-22] and sign-extended to 32-bits. This is needed for instructions: LD, ST, SVPC, INC, as Ill as for computing branch and jump targets.

Control Unit

The control unit interprets instruction opcode bits [3:0] and produces WB, M, and EX signals. Refer to Table 4 : Signal Size for more information.

Execute (EX) Stage 3

This stage contains the following component:

- ALU
- adder
- ALU Control
- mux

ALU and ALU Control

The ALU implements the operands listed in Table 2 : ALU Truth Table.
The ALUOp (2-bit) signal from EX feeds into the ALU control unit which provides the correct (add, sub, neg) signals to the ALU operation.

Branch and Jump Addressing

The adder computes PC + y for SVPC.
For branching, the address stored in rs_data is referred by the MEM stage.
For jumping, I use the rs_data that is referred in the MEM stage.

Data Memory (MEM) Stage 4

This stage contains the following components:

- Data Memory
- AND and OR gates

Data Memory takes two signals:

- MemRead: $x\text{rs} \leftarrow \text{MEM}[x\text{rs} + y]$
- MemWrite: $\text{MEM}[x\text{rs} + y] \leftarrow \text{rt}$

The AND and OR gates are used to calculate PCSrc, simply:

$$\text{PCSrc} = \text{N_MEM} \ \& \ \text{N_EX} \mid \text{Z_MEM} \ \& \ \text{Z_EX}$$

Write-Back (WB) Stage 5

This stage contains the following components:

- mux: two that are connected to each other

These mux are connected and controlled as follows:

1. Signal: MemToReg; 0 – OUT ; 1 – read_data
2. Signal: SVPC ; 0 – 1st mux; 1 – PC_imm

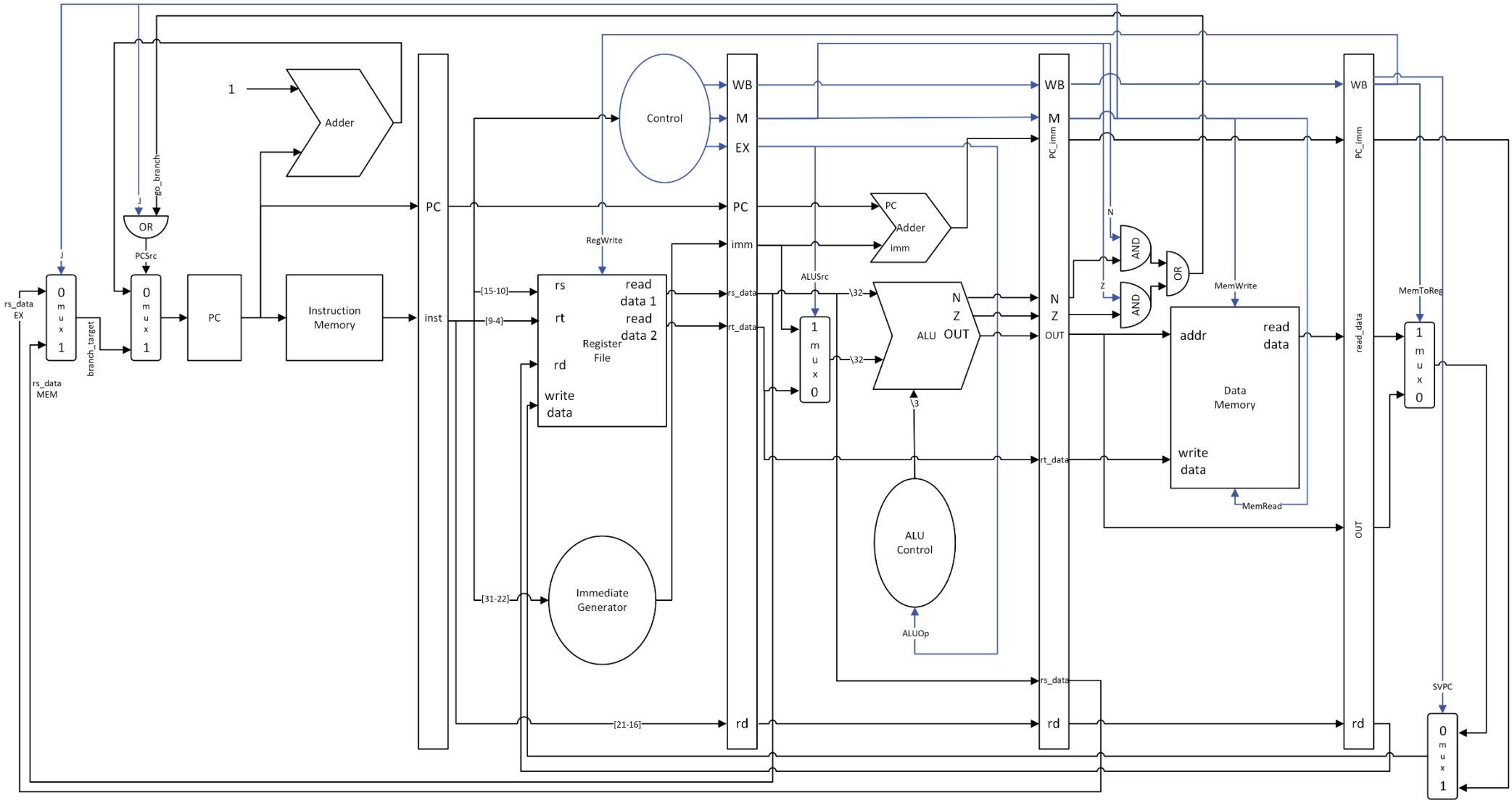
OUT is the ALU result.
read_data is from the data memory.
PC_imm is simply PC + immediate value.
I

Instruction Support

Instruction	Syntax	y	rd	rs	rt	Opcode	Function
No operation	NOP	x	x	x	x	0000	
Save PC	SVPC rd, y	y	rd	x	x	1111	
Load	LD rd, rs, y	y	rd	rs	x	1110	
Store	ST rt, rs, y	y	x	rs	rt	0011	
Add	ADD rd, rs, rt	x	rd	rs	rt	0100	
Increment	INC rd, rs, y	y	rd	rs	x	0101	
Negate	NEG rd, rs	x	rd	rs	x	0110	
Subtract	SUB rd, rs, rt	x	rd	rs	rt	0111	
Jump	J rs	x	x	rs	x	1000	
Branch if zero	BRZ rs	x	x	rs	x	1001	
Branch if negative	BRN rs	x	x	rs	x	1010	

Special instruction SCU ISA supports are NOP, SVPC, BRN, and BRZ.
NOP will deactivate all control signal and is used as a bubble.
SVPC saves the return location when branching. It can also be used to refer specific functions.
BRN will branch if the previous instruction calculated a negative result in the ALU.
BRZ will branch if the previous instruction calculated a zero result in the

Datapath Diagram



Simulation Results

```
Starting Simulation.
--------------------------------------------------
Checking A array is correctly initialized
--------------------------------------------------
[PASS] Index x2: Correct (3)
[PASS] Index x3: Correct (1)
[PASS] Index x4: Correct (5)
[PASS] Index x5: Correct (7)
[PASS] Index x6: Correct (2)
[PASS] Index x7: Correct (9)
[PASS] Index x8: Correct (8)
--------------------------------------------------
Checking SVPC Data Setup (<00>-<07>)
--------------------------------------------------
[PASS] <00> x40: Correct immediate (19)
[PASS] <01> x41: Correct immediate (70)
[PASS] <02> x42: Correct immediate (50)
[PASS] <03> x43: Correct immediate (56)
[PASS] <04> x44: Correct immediate (63)
[PASS] <05> x45: Correct immediate (30)
[PASS] <06> x46: Correct immediate (34)
[PASS] <07> x47: Correct immediate (38)
--------------------------------------------------
Checking Final Result (Data Mem - Indices 71-77)
--------------------------------------------------
[PASS] Index x71: Correct (3)
[PASS] Index x72: Correct (3)
[PASS] Index x73: Correct (5)
[PASS] Index x74: Correct (5)
[PASS] Index x75: Correct (7)
[PASS] Index x76: Correct (8)
[PASS] Index x77: Correct (8)
--------------------------------------------------
```

Figure 1: Simulation Result

Simulation Report

The simulation code (Code 1) I ran resulted in the correct initialization and results as observed in Figure 1.

Performance Analysis

The Median 1D Stencil code that was edited with my NOP instruction took **2510 ns** to complete with a clock of **100 MHz**.

$$f = \frac{1}{T} \text{where } T = 10 \text{ ns}$$

$$= \frac{1}{10 \cdot 10^{-9}} = 100 \cdot 10^{6} \text{ hz} \equiv 100 \text{ Mhz}$$

This also means that it <u>took 251 instructions to complete</u> with an array of $\boldsymbol{n} = \mathbf{7}$.

$$\text{CPU time} = \text{Cycle Time} \cdot \text{Cycle Count}$$

$$\text{Cycle Count} = \frac{\text{CPU time}}{\text{Cycle Time}} = \frac{2510 \cdot 10^{-9}}{10 \cdot 10^{-9}} = 251 \text{ cycles}$$

Since the program executed 251 instructions for an input array of $n = 7$, the CPI estimate is:

$$\text{CPI} \approx \frac{\text{Total Cycles}}{\text{Instruction Count}} = \frac{251}{251} \approx 1.0$$

This is because I didn't install hazard unit and the inserted NOPs correspond to the added instruction.

To verify this I can estimate the number of looped and non-looped instruction.

$$\text{Instruction Count}(n) = I_{\text{non-loop}} + I_{\text{loop}} \cdot n$$

$$\text{Cycle Count}(n) \approx I_{\text{non-loop}} + I_{\text{loop}} \cdot n$$

For simulation of $n = 7$

$$I_{\text{loop}} \approx \frac{251 - I_{\text{non-loop}}}{7} \approx \frac{251 - 36}{7} \approx 30 \text{ instructions per iteration}$$

This verification lines up with
Verification: (Final Result) and **Waveform** Example 1.
I could further reduce the number of cycles by implementing a forwarding unit or hazard detection but that was not done which caused us to add additional NOP instructions into my program.

Appendix

Instruction Truth Table:

Inst.	OP	ALUSrc	MemToReg	RegWrite	MemRead	MemWrite	ALUOp (2)	SVPC	Z	N	J
NOP	0000	0	0	0	0	0	xx	0	0	0	0
SVPC	1111	1	x	1	0	0	00	1	0	0	0
LD	1110	1	1	1	1	0	00	0	0	0	0
ST	0011	1	0	0	0	1	00	0	0	0	0
ADD	0100	0	0	1	0	0	00	0	0	0	0
INC	0101	1	0	1	0	0	00	0	0	0	0
NEG	0110	0	0	1	0	0	10	0	0	0	0
SUB	0111	0	0	1	0	0	01	0	0	0	0
J	1000	0	x	0	0	0	xx	x	x	x	1
BRZ	1001	x	x	0	0	0	xx	x	1	0	0
BRN	1010	x	x	0	0	0	xx	x	0	1	0

Table 1: Instruction Truth Table

ALU Truth Table

ALUOp	Instruction [3:0]	ALU Control Output (3 bits)	Operation
00	1111,1110,0100,0101,0011	000	Add
01	0111,1001	101	Subtract
10	0110	110	Negate
11			Pass A

Table 2 : ALU Truth Table

ALU Control Output Signals: ADD, NEG, SUB

ALU Truth Table:

ADD	NEG	SUB	OUT	Operation
0	0	0	A + B	Add
0	1	0	Don't Care	No Operation
1	0	1	A – B	Subtract
1	1	0	-A	2's Complement
1	1	1	A	Pass A

Table 3: ALU Unit Truth Table

Signals:

WB	(3 bits total):	SVPC, MemToReg, RegWrite
IF/EX	(5 bits total):	J, N, Z, MemRead, MemWrite
EX/MEM	(3 bits total):	ALU Op (2 bits), ALUSrc

Table 4 : Signal Size

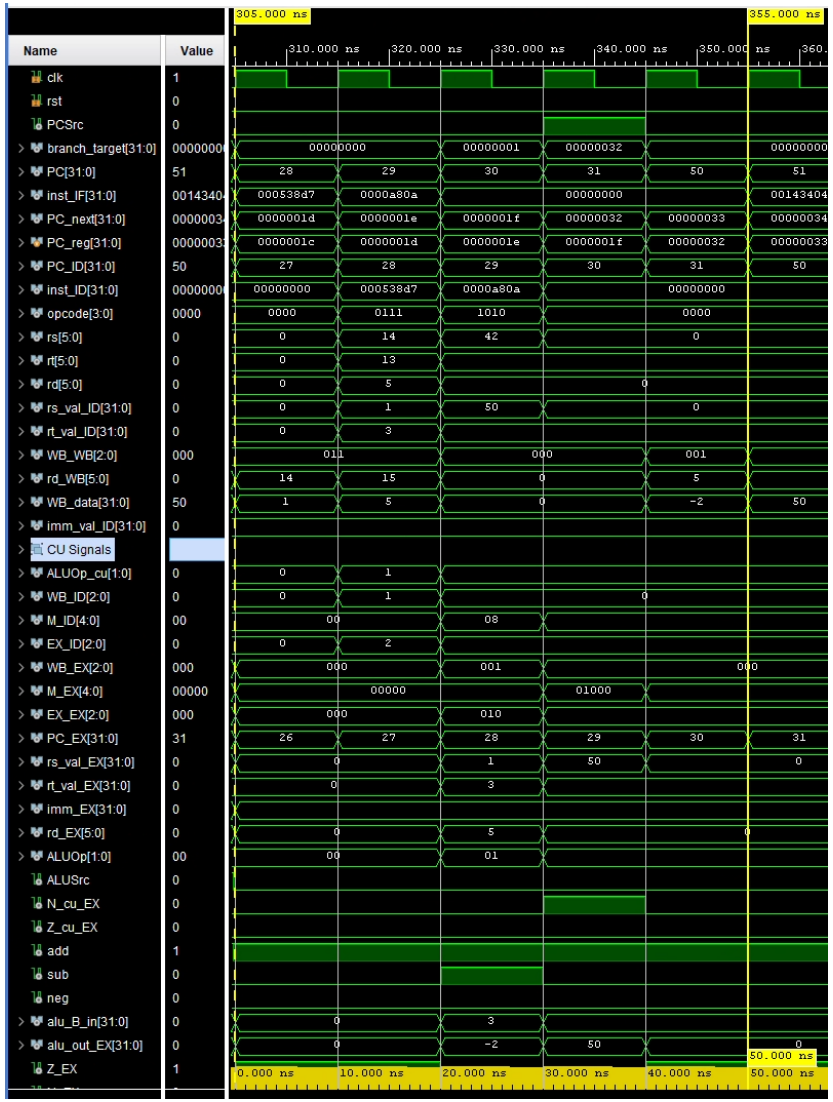
Buffer Size:

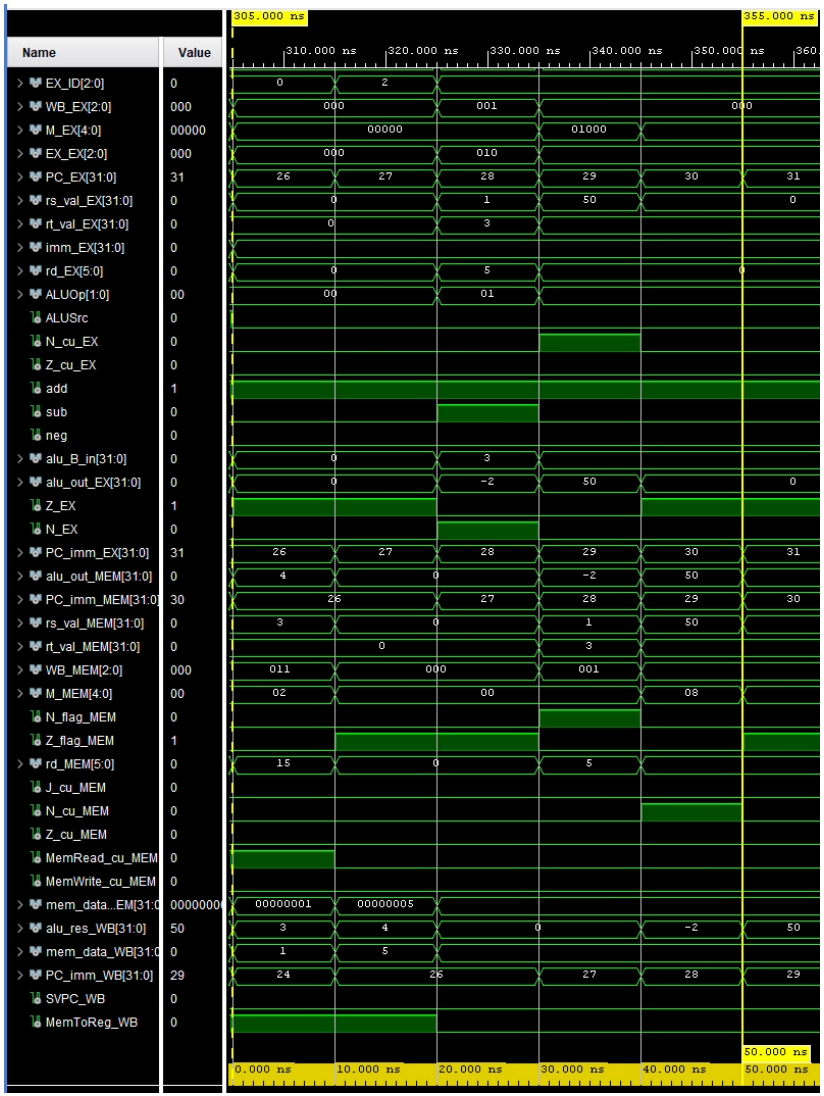
ID/IF	(64 bits total):	PC(32), inst(32)
IF/EX	(145 bits total):	WB(3), M(5), EX(3), PC(32), imm(32), rs_data(32), rt_data(32), rd(6)
EX/MEM	(144 bits total):	WB(3), M(5), PC_imm(32), N(1), Z(1), OUT(32), rs_data(32), rt_data(32), rd(6)
MEM/WB	(105 bits total):	WB(3), PC_imm(32), read_data(32), OUT(32), rd(6)

Table 5 : Buffer Size

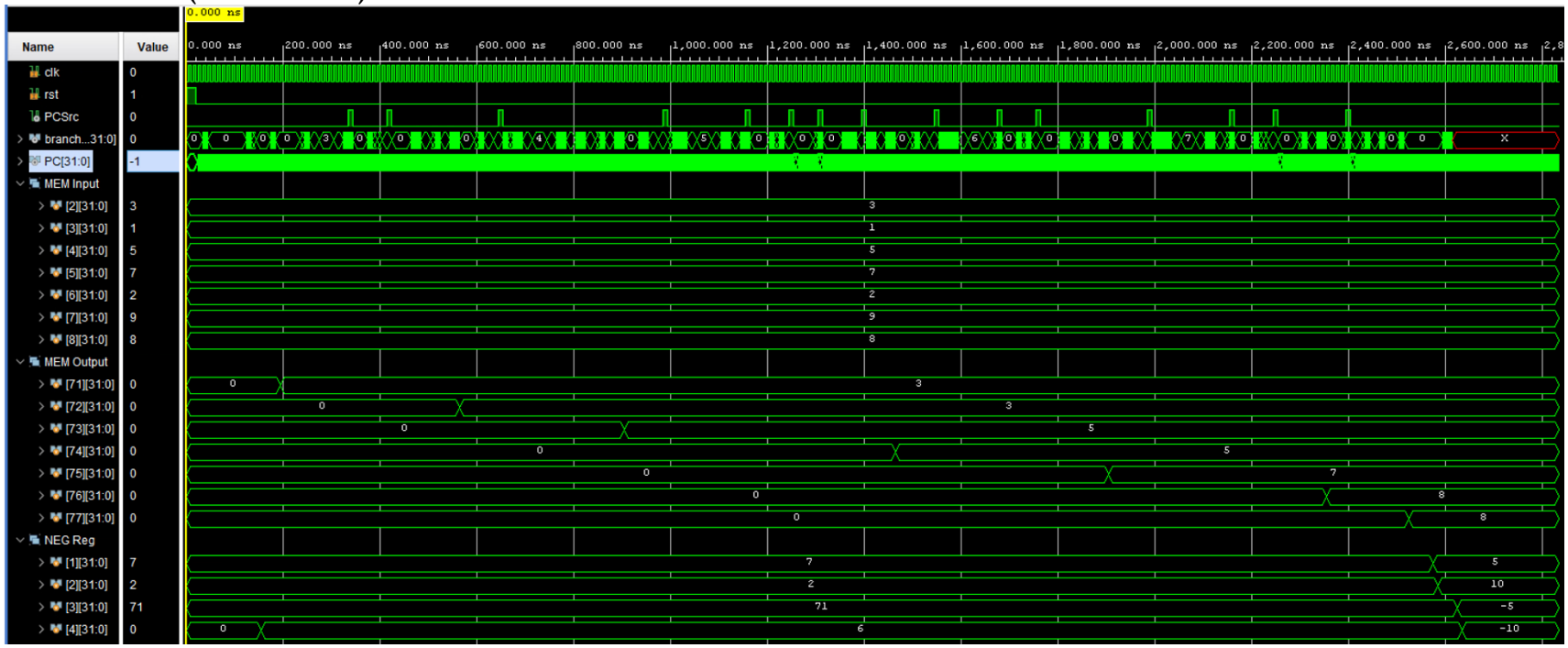
Note that buffer has CLK and RST signal, adding an additional 2 (that is not included in calculation above).

Waveform Example 1





Verification: (Final Result)



I can visually see that MEM Input was set correctly since the beginning of the program, labeled as `**MEM Input**`. As the program progresses and finds the median, I can visually see `**MEM Output**` getting set individually. I verified that this result is correct, as I set my array `**B**` to be in the data memory address `**71**`.
The median stencil code that was provided by the professor did not have NEG instruction so I tested it at the end of the median stencil code. I set registry `**1**` to 5 and registry `**2**` to 10. I then made sure that running the negative instruction would store the negative values into the registry (as observed in registry `**3**` and `**4**`).
Refer to Code 5 for more the smyce code of NEG test.

Verification: each loop is around 30 instructions, if you measure the difference betIen the `**MEM Output**` updates.
e.g., the difference betIen data memory address `**73**` and `**72**` is around 320ns which is 32 cycles and with CPI of 1.0, that's 32 instructions.

Code 1: Simulation (tb_cpu.v)

```
`timescale 1ns / 1ps
module tb_cpu;
    reg clk;
    reg rst;

    cpu uut (
        .clk(clk),
        .rst(rst)
    );

    initial begin
        clk = 0;
        forever #5 clk = ~clk;
    end

    initial begin
        rst = 1;
        #20;
        rst = 0;
        $display("Starting Simulation.");
    end

    integer i;
    reg [6:0]  idx [0:7];
    reg [31:0] exp_val [0:7];

    initial begin
        $display("----------------------------------------------------");
        $display("Checking A array is correctly initialized         ");
        $display("----------------------------------------------------");


        exp_val[0] = 32'd3;
        exp_val[1] = 32'd1;
        exp_val[2] = 32'd5;
        exp_val[3] = 32'd7;
        exp_val[4] = 32'd2;
        exp_val[5] = 32'd9;
        exp_val[6] = 32'd8;
    end

    initial begin
        #20;
        #80;
        #50;

        $display("----------------------------------------------------");
        $display("Checking SVPC Data Setup (<00>-<07>)              ");
        $display("----------------------------------------------------");

        idx[0] = 7'd40;  exp_val[0] = 32'd19;
        idx[1] = 7'd41;  exp_val[1] = 32'd70;
        idx[2] = 7'd42;  exp_val[2] = 32'd50;
        idx[3] = 7'd43;  exp_val[3] = 32'd56;
        idx[4] = 7'd44;  exp_val[4] = 32'd63;
        idx[5] = 7'd45;  exp_val[5] = 32'd30;
        idx[6] = 7'd46;  exp_val[6] = 32'd34;
        idx[7] = 7'd47;  exp_val[7] = 32'd38;

        for (i = 0; i < 8; i = i + 1) begin
            if (uut.regfile.regs[idx[i]] === exp_val[i]) begin
                $display("[PASS] <%02d> x%0d: Correct immediate (%0d)",
                        i, idx[i], exp_val[i]);
            end else begin
                $display("[FAIL] <%02d> x%0d: Found %0d, Expected (%0d)",
                        i, idx[i], uut.regfile.regs[idx[i]], exp_val[i]);
            end
        end
    end

    initial begin
        #3500;
        idx[0] = 7'd71; exp_val[0] = 32'd3;
        idx[1] = 7'd72; exp_val[1] = 32'd3;
        idx[2] = 7'd73; exp_val[2] = 32'd5;
        idx[3] = 7'd74; exp_val[3] = 32'd5;
        idx[4] = 7'd75; exp_val[4] = 32'd7;
        idx[5] = 7'd76; exp_val[5] = 32'd8;
        idx[6] = 7'd77; exp_val[6] = 32'd8;

        $display("----------------------------------------------------");
        $display("Checking Final Result (Data Mem - Indices 71-77)  ");
        $display("----------------------------------------------------");
        for (i = 0; i < 7; i = i + 1) begin
            if (uut.dm.mem[idx[i]] === exp_val[i]) begin
                $display("[PASS] Index x%0d: Correct (%0d)",
                        idx[i], exp_val[i]);
            end else begin
                $display("[FAIL] Index x%0d: Found %0d, Expected (%0d)",
                        idx[i], uut.dm.mem[idx[i]], exp_val[i]);
            end
        end

        $display("----------------------------------------------------");
        $finish;
    end

endmodule
```

Code 2: JavaScript Convert Code (convert.js)

```javascript
const COMMENT_DELIM = "//";
const fs = require('fs');
const readline = require('readline').createInterface({
    input: process.stdin,
    output: process.stdout
});
readline.question('Enter the path to the assembly file: ', filePath => {
    fs.readFile(filePath, 'utf8', (err, data) => {
        if (err) {
            console.error('Error reading file:', err);
            readline.close();
            return;
        }
        const lines = data.split('\n');
        let outputLines = [];
        for (let line of lines) {
            if (line.includes(COMMENT_DELIM))
                line = line.split(COMMENT_DELIM)[0].trim();
            if (line === '') continue;
            line = line.replace(/\s+/g, ' ');
            const match = line.match(/<(\d+)>:\s+([A-Z]+)(?:\s+([x0-9-,\s]+))?/);
            if (match) {
                const address = parseInt(match[1]);
                const opcode = match[2];
                let operands = undefined;
                if (match[3])
                    operands = match[3].split(',').map(op => op.trim());
                const binaryInstruction = convertToBinary(opcode, operands);
                outputLines.push(`memory[${address}] = 32'b${binaryInstruction};`);
            }
        }
        const outputFilePath = filePath.replace('.asm', '_convert.txt');
        fs.writeFile(outputFilePath, outputLines.join('\n'), err => {
            if (err) console.error('Error writing file:', err);
            else console.log('File converted successfully:', outputFilePath);
            readline.close();
        });
    });
});
function convertToBinary(opcode, operands) {
    const opcodeMap = {
        'NOP': '0000',
        'SVPC': '1111',
        'LD': '1110',
        'ST': '0011',
        'ADD': '0100',
        'INC': '0101',
        'NEG': '0110',
        'SUB': '0111',
        'J': '1000',
        'BRZ': '1001',
        'BRN': '1010'
    };
    let Y = '0000000000';
    let RD = '000000';
    let RS = '000000';
    let RT = '000000';
    let OPCODE = opcodeMap[opcode];
    switch (opcode) {
        case 'NOP':
            break;
        case 'SVPC':
            RD = registerToBinary(operands[0]);
            Y = immediateToBinary(operands[1], 10);
            break;
        case 'LD':
            RD = registerToBinary(operands[0]);
            RS = registerToBinary(operands[1]);
            Y = immediateToBinary(operands[2], 10);
            break;
        case 'ST':
            RT = registerToBinary(operands[0]);
            RS = registerToBinary(operands[1]);
            Y = immediateToBinary(operands[2], 10);
            break;
        case 'ADD':
            RD = registerToBinary(operands[0]);
            RS = registerToBinary(operands[1]);
            RT = registerToBinary(operands[2]);
            break;
        case 'INC':
            RD = registerToBinary(operands[0]);
            RS = registerToBinary(operands[1]);
            Y = immediateToBinary(operands[2], 10);
            break;
        case 'NEG':
            RD = registerToBinary(operands[0]);
            RS = registerToBinary(operands[1]);
            break;
        case 'SUB':
            RD = registerToBinary(operands[0]);
            RS = registerToBinary(operands[1]);
            RT = registerToBinary(operands[2]);
            break;
        case 'J':
            RS = registerToBinary(operands[0])
            break;
        case 'BRZ':
            RS = registerToBinary(operands[0])
            break;
        case 'BRN':
            RS = registerToBinary(operands[0])
            break;
        default:
            console.error('Unknown opcode:', opcode);
            break;
    }
    return `${Y}_${RD}_${RS}_${RT}_${OPCODE}`;
}

function registerToBinary(register) {
    const regNum = parseInt(register.slice(1));
    return regNum.toString(2).padStart(6, '0');
}

function immediateToBinary(immediate, bits) {
    let immNum = parseInt(immediate);
    if (immNum < 0) {
        immNum = (1 << bits) + immNum;
    }
    return immNum.toString(2).padStart(bits, '0');
}
```

Code 3: ASM Code (Median_stencil.asm)

```
<00>:  SVPC   x40, 19        // save addr of LOOP_BODY (y = 19-0)
<01>:  SVPC   x41, 69        // save addr of EPILOGUE (y = 70-1)
<02>:  SVPC   x42, 48        // save addr of SWAP_1 (y = 50-2)
<03>:  SVPC   x43, 53        // save addr of SWAP_2 (y = 56-3)
<04>:  SVPC   x44, 59        // save addr of SWAP_3 (y = 63-4)
<05>:  SVPC   x45, 25        // return addr from SWAP_1 (y = 30-5)
<06>:  SVPC   x46, 28        // return addr from SWAP_2 (y = 34-6)
<07>:  SVPC   x47, 31        // return addr from SWAP_3 (y = 38-7)
<08>:  INC    x4, x1, -1     // save n-1 to x4
<09>:  LD     x20, x2, 0     // load A[0] to x20;  y=0, rd=x10, rs=x2, rt=NA
<10>:  NOP
<11>:  NOP
<12>:  NOP
<13>:  ST     x20, x3, 0     // store x20 to B[0]; y=0, rd=NA, rs=x3, rt=x10
<14>:  SUB    x0, x0, x0     // making x0 as 0
<15>:  INC    x10, x0, 1     // set x10 (for variable i) as 1
<16>:  NOP
<17>:  NOP
<18>:  NOP
// LOOP_BODY:
<19>:  ADD    x11, x2, x10   // addr of A[i]:x11=&A[0]+i
<20>:  ADD    x12, x3, x10   // addr of B[i]:x12=&B[0]+i
<21>:  NOP
<22>:  NOP
<23>:  LD     x13, x11, -1   // load A[i-1] to x13
<24>:  LD     x14, x11, 0    // load A[i] to x14
<25>:  LD     x15, x11, 1    // load A[i+1] to x15
<26>:  NOP
<27>:  NOP
<28>:  SUB    x5, x14, x13   // x5 = x14 - x13 = 1 - 3 = -2 (correct output)
<29>:  BRN    x42            // SWAP_1 if x13 > x14 : x42 (got address 34)
<30>:  NOP                   // SWAP_1 returns here to avoid hazard
<31>:  NOP
<32>:  SUB    x5, x15, x14
<33>:  BRN    x43            // SWAP_2 if x15 > x14
<34>:  NOP                   // SWAP_2 returns here to avoid hazard
<35>:  NOP
<36>:  SUB    x5, x14, x13
<37>:  BRN    x44            // SWAP_3 if x13 > x14
                             // after this, x14 holds the median of
                             // the three elements
<38>:  NOP                   // SWAP_3 returns here to avoid hazard
<39>:  NOP
<40>:  ST     x14, x12, 0    // save x14 to B[i]
<41>:  INC    x10, x10, 1    // i = i+1
<42>:  NOP
<43>:  NOP
<44>:  NOP
<45>:  SUB    x5, x10, x4    // to compare i and n-1
<46>:  BRZ    x41            // goto epilogue if epilogue if i == n-1
<47>:  NOP
<48>:  NOP
<49>:  J      x40            // repeat LOOP_BODY
// SWAP_1 (swap x13 and x14)
<50>:  NOP
<51>:  ADD    x20, x13, x0   // x13 = 3
<52>:  ADD    x13, x14, x0   // x14 = 1
<53>:  NOP
<54>:  J      x45
<55>:  ADD    x14, x20, x0   // x14 = 3
// SWAP_2 (swap x14 and x15)
<56>:  NOP
<57>:  NOP
<58>:  ADD    x20, x14, x0
<59>:  ADD    x14, x15, x0
<60>:  NOP
<61>:  J      x46
<62>:  ADD    x15, x20, x0
// SWAP_3 (swap x13 and x14)
<63>:  NOP
<64>:  NOP
<65>:  ADD    x20, x13, x0
<66>:  ADD    x13, x14, x0
<67>:  NOP
<68>:  J      x47
<69>:  ADD    x14, x20, x0
// EPILOGUE
<70>:  ADD    x11, x2, x10
<71>:  ADD    x12, x3, x10
<72>:  NOP
<73>:  NOP
<74>:  LD     x20, x11, 0
<75>:  NOP
<76>:  NOP
<77>:  NOP
<78>:  ST     x20, x12, 0
<79>:  NOP
<80>:  NOP
<81>:  SUB    x0, x0, x0
<82>:  INC    x1, x0, 5
<83>:  INC    x2, x0, 10
<84>:  NEG    x3, x1
<85>:  NEG    x4, x2
// For next instruction (technically not needed but added for safety)
<86>:  NOP
<87>:  NOP
<88>:  NOP
```

Code 4: ASM Median Stencil Converted (Binary Program)

```
memory[0] = 32'b0000010011_101000_000000_000000_1111;
memory[1] = 32'b0001000101_101001_000000_000000_1111;
memory[2] = 32'b0000110000_101010_000000_000000_1111;
memory[3] = 32'b0000110101_101011_000000_000000_1111;
memory[4] = 32'b0000111011_101100_000000_000000_1111;
memory[5] = 32'b0000011001_101101_000000_000000_1111;
memory[6] = 32'b0000011100_101110_000000_000000_1111;
memory[7] = 32'b0000011111_101111_000000_000000_1111;
memory[8] = 32'b1111111111_000100_000001_000000_0101;
memory[9] = 32'b0000000000_010100_000010_000000_1110;
memory[10] = 32'b0000000000_000000_000000_000000_0000;
memory[11] = 32'b0000000000_000000_000000_000000_0000;
memory[12] = 32'b0000000000_000000_000000_000000_0000;
memory[13] = 32'b0000000000_000000_000011_010100_0011;
memory[14] = 32'b0000000000_000000_000000_000000_0111;
memory[15] = 32'b0000000001_001010_000000_000000_0101;
memory[16] = 32'b0000000000_000000_000000_000000_0000;
memory[17] = 32'b0000000000_000000_000000_000000_0000;
memory[18] = 32'b0000000000_000000_000000_000000_0000;
memory[19] = 32'b0000000000_001011_000010_001010_0100;
memory[20] = 32'b0000000000_001100_000011_001010_0100;
memory[21] = 32'b0000000000_000000_000000_000000_0000;
memory[22] = 32'b0000000000_000000_000000_000000_0000;
memory[23] = 32'b1111111111_001101_001011_000000_1110;
memory[24] = 32'b0000000000_001110_001011_000000_1110;
memory[25] = 32'b0000000001_001111_001011_000000_1110;
memory[26] = 32'b0000000000_000000_000000_000000_0000;
memory[27] = 32'b0000000000_000000_000000_000000_0000;
memory[28] = 32'b0000000000_000101_001110_001101_0111;
memory[29] = 32'b0000000000_000000_101010_000000_1010;
memory[30] = 32'b0000000000_000000_000000_000000_0000;
memory[31] = 32'b0000000000_000000_000000_000000_0000;
memory[32] = 32'b0000000000_000101_001111_001110_0111;
memory[33] = 32'b0000000000_000000_101011_000000_1010;
memory[34] = 32'b0000000000_000000_000000_000000_0000;
memory[35] = 32'b0000000000_000000_000000_000000_0000;
memory[36] = 32'b0000000000_000101_001110_001101_0111;
memory[37] = 32'b0000000000_000000_101100_000000_1010;
memory[38] = 32'b0000000000_000000_000000_000000_0000;
memory[39] = 32'b0000000000_000000_000000_000000_0000;
memory[40] = 32'b0000000000_000000_001100_001110_0011;
memory[41] = 32'b0000000001_001010_001010_000000_0101;
memory[42] = 32'b0000000000_000000_000000_000000_0000;
memory[43] = 32'b0000000000_000000_000000_000000_0000;
memory[44] = 32'b0000000000_000000_000000_000000_0000;
memory[45] = 32'b0000000000_000101_001010_000100_0111;
memory[46] = 32'b0000000000_000000_101001_000000_1001;
memory[47] = 32'b0000000000_000000_000000_000000_0000;
memory[48] = 32'b0000000000_000000_000000_000000_0000;
memory[49] = 32'b0000000000_000000_101000_000000_1000;
memory[50] = 32'b0000000000_000000_000000_000000_0000;
memory[51] = 32'b0000000000_010100_001101_000000_0100;
memory[52] = 32'b0000000000_001101_001110_000000_0100;
memory[53] = 32'b0000000000_000000_000000_000000_0000;
memory[54] = 32'b0000000000_000000_101101_000000_1000;
memory[55] = 32'b0000000000_001110_010100_000000_0100;
memory[56] = 32'b0000000000_000000_000000_000000_0000;
memory[57] = 32'b0000000000_000000_000000_000000_0000;
memory[58] = 32'b0000000000_010100_001110_000000_0100;
memory[59] = 32'b0000000000_001110_001111_000000_0100;
memory[60] = 32'b0000000000_000000_000000_000000_0000;
memory[61] = 32'b0000000000_000000_101110_000000_1000;
memory[62] = 32'b0000000000_001111_010100_000000_0100;
memory[63] = 32'b0000000000_000000_000000_000000_0000;
memory[64] = 32'b0000000000_000000_000000_000000_0000;
memory[65] = 32'b0000000000_010100_001101_000000_0100;
memory[66] = 32'b0000000000_001101_001110_000000_0100;
memory[67] = 32'b0000000000_000000_000000_000000_0000;
memory[68] = 32'b0000000000_000000_101111_000000_1000;
memory[69] = 32'b0000000000_001110_010100_000000_0100;
memory[70] = 32'b0000000000_001011_000010_001010_0100;
memory[71] = 32'b0000000000_001100_000011_001010_0100;
memory[72] = 32'b0000000000_000000_000000_000000_0000;
memory[73] = 32'b0000000000_000000_000000_000000_0000;
memory[74] = 32'b0000000000_010100_001011_000000_1110;
memory[75] = 32'b0000000000_000000_000000_000000_0000;
memory[76] = 32'b0000000000_000000_000000_000000_0000;
memory[77] = 32'b0000000000_000000_000000_000000_0000;
memory[78] = 32'b0000000000_000000_001100_010100_0011;
memory[79] = 32'b0000000000_000000_000000_000000_0000;
memory[80] = 32'b0000000000_000000_000000_000000_0000;
memory[81] = 32'b0000000000_000000_000000_000000_0111;
memory[82] = 32'b0000000101_000001_000000_000000_0101;
memory[83] = 32'b0000001010_000010_000000_000000_0101;
memory[84] = 32'b0000000000_000011_000001_000000_0110;
memory[85] = 32'b0000000000_000100_000010_000000_0110;
memory[86] = 32'b0000000000_000000_000000_000000_0000;
memory[87] = 32'b0000000000_000000_000000_000000_0000;
memory[88] = 32'b0000000000_000000_000000_000000_0000;
```

Code 5: NEG Instruction (Continuation of the previous median program)

```
<79>:    NOP
<80>:    NOP
<81>:    SUB     x0, x0, x0
<82>:    INC     x1, x0, 5
<83>:    INC     x2, x0, 10
<84>:    NOP
<85>:    NOP
<86>:    NOP
<87>:    NEG     x3, x1
<88>:    NEG     x4, x2
```

Code 6 : NEG Binary

```
memory[79] = 32'b0000000000_000000_000000_000000_0000;
memory[80] = 32'b0000000000_000000_000000_000000_0000;
memory[81] = 32'b0000000000_000000_000000_000000_0111;
memory[82] = 32'b0000000101_000001_000000_000000_0101;
memory[83] = 32'b0000001010_000010_000000_000000_0101;
memory[84] = 32'b0000000000_000000_000000_000000_0000;
memory[85] = 32'b0000000000_000000_000000_000000_0000;
memory[86] = 32'b0000000000_000000_000000_000000_0000;
memory[87] = 32'b0000000000_000011_000001_000000_0110;
memory[88] = 32'b0000000000_000100_000010_000000_0110;
```